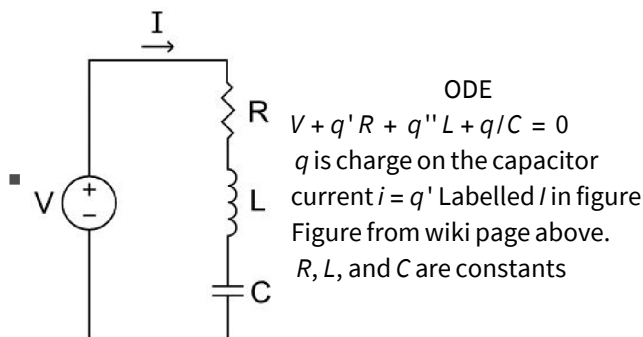


Name

M#

Models

- Mixing in Tanks <https://xkcd.com/2974/>
 - $y' = Ay + f$ with $y(0) = y_0$ given.
 - Conservation of volume v and salt s for each tank
 - $v'(t) = \text{rate in} - \text{rate out}$
 - $s'(t) = \text{rate in} - \text{rate out}$
- Probes in Earth-Moon system. CR3BP <https://orbital-mechanics.space/the-n-body-problem/circular-restricted-three-body-problem.html>
 - With $M = 1/82.45$ and $E = 1 - M$ and $r_1(x, y) = \sqrt{(x + M)^2 + y^2}$ and $r_2 = \sqrt{(x - E)^2 + y^2}$
$$\begin{aligned}x'' &= 2y' + x - \frac{E(x+M)}{r_1^3} - \frac{M(x-E)}{r_2^3} \\ y'' &= -2x' + y - \frac{Ey}{r_1^3} - \frac{My}{r_2^3}\end{aligned}$$
with $y(0), x(0), x'(0), y'(0)$ given
 - Follows from Newton's law of gravitation in a rotating coordinate system.
- RLC circuits https://en.wikipedia.org/wiki/RLC_circuit
 - Circuit Rules: https://en.wikipedia.org/wiki/Kirchhoff%27s_circuit_laws
 - Current i is the rate of change of charge q (usually on a capacitor) through a wire.
 - Voltage drops due to resistors ($V_R = iR$), capacitors ($V_C = \frac{q}{C}$), inductors ($V_L = L di/dt$), and voltage sources sum to zero around a circuit loop.
 - Currents sum to zero at a junction.



- Population models https://en.wikipedia.org/wiki/Population_model
 - Growth of population p
 - $p'(t) = \text{birth rate} - \text{death rate}$

- Estimate death rate using life expectancy. estimate birth rate using litter size etc.
- Predator (wolf w) prey (moose m) https://en.wikipedia.org/wiki/Lotka%E2%80%93Volterra_equations
 - $m'(t)$ = birth rate – death rate (Assume moose death rate proportional to wolf and moose population)
 - $w'(t)$ = birth rate – death rate (Assume wolf birth rate proportional to wolf and moose population)
 - Lotka-Volterra Model

$$\begin{aligned} m' &= \alpha m - \beta m w \\ w' &= -\gamma w + \delta m w \end{aligned}$$

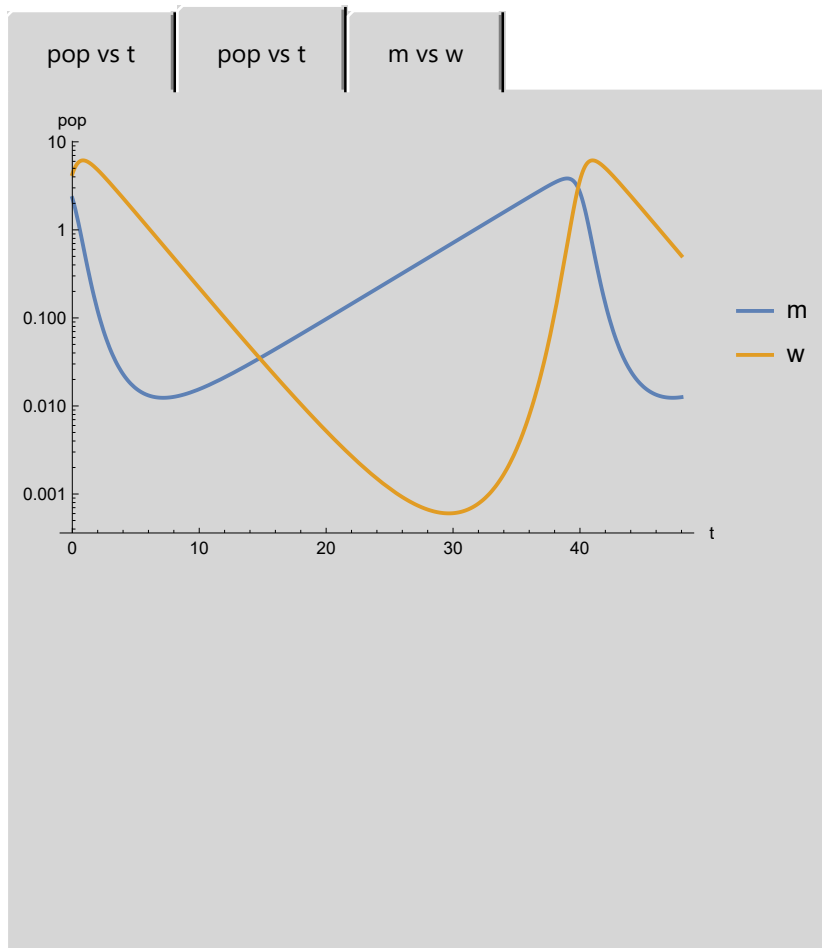
```

In[ ]:= {α, β, γ, δ} = {0.2, 0.3, 0.4, 0.6};
TMax = 48;
{w0, m0} = {4.3, 2.3};
{mSol, wSol} = NDSolveValue[{
  m'[t] == α m[t] - β m[t] * w[t], m[0] == m0,
  w'[t] == -γ w[t] + δ m[t] * w[t], w[0] == w0
}, {m, w}, {t, 0, TMax}];

TabView[{
  "pop vs t" → Plot[{mSol[t], wSol[t]},
    {t, 0, TMax}, AxesLabel → {"t", "pop"}, PlotLegends → {"m", "w"}],
  "pop vs t" → LogPlot[{mSol[t], wSol[t]},
    {t, 0, TMax}, AxesLabel → {"t", "pop"}, PlotLegends → {"m", "w"}],
  "m vs w" → ParametricPlot[{mSol[t], wSol[t]},
    {t, 0, TMax}, AxesLabel → {"m", "w"}, PlotRange → All]
}]

```

Out[]:=



First order ODE: Analytical Solutions

- A separable scalar ODE has the form $y'(t) = g(t)h(y)$.

You solved these in Calculus II by separating variables and integrating

$$dy/dt = g(t)h(y) \implies \int \frac{dy}{h(y)} = \int g(t) dt + C$$

An initial condition $y(0) = y_0$ would let you solve for the one constant of integration.

- A linear 1st order ODE has the standard form $y'(t) + p(t)y(t) = f(t)$. Solve by computing the IF $v(t) = e^{\int_0^t p(\tau) d\tau}$ and solve the separable ODE $(vy)' = v f$.

- A 1st order ODE of the form $\frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial y} dy = 0$ is exact with general solution $f(t, y) = C$.

- $M(t, y) dt + N(t, y) dy = 0$ is exact if and only if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$

- To solve compute $f(t, y) = \int M(t, y) dt + h(y) \implies h'(y) = N(t, y) - \frac{\partial}{\partial y} [\int M(t, y) dt]$ then integrate $h'(y)$ and write down the general solution $f(t, y) = C$

- 2.7 Substitution. Some ODEs look nicer in other variables. For example

- 2.7.4 Bernoulli: $y' + p(t)y = q(t)y^n$. The substitution $u = y^{1-n}$ gives a linear equation in u .

- $u' = (1-n)y^{1-n-1}y' = (1-n)y^{-n}y'$

- $(1-n)y^{-n}y' + p(t)(1-n)y^{-n}y = q(t)(1-n)y^{-n}y^n$

- $u' + (1-n)p(t)u = (1-n)q(t)$ A linear ODE we know how to solve.

- Just like in Calculus II you need to translate every y into u .

- Other substitutions are possible. I could ask you to make a substitution on an exam.